Raspberry Pi

Face & Voice Recognition + Health Monitor

Complete Step-by-Step Project Guide

Project Overview

Component	Details
Hardware	Raspberry Pi 4B, USB Webcam, USB Microphone
Sensors	MAX30102 Heart Rate, NEO-M8N GPS, 1.3" OLED Display
People	Bakiya, Ranjani, Srikavi, Swetha
Features	Face recognition, Voice recognition, Health monitoring, GPS tracking
Platform	Windows PC (training) + Raspberry Pi 4B (deployment)

SECTION 1: Windows PC Setup & Model Training

All AI model training is done on Windows PC first, then transferred to Raspberry Pi.

1.1 Install Python Packages on Windows

Open PyCharm terminal and run:

```
pip install deepface tf-keras
pip install opencv-python opencv-contrib-python
pip install librosa scikit-learn sounddevice numpy
```

NOTE: opencv-contrib-python is required for LBPH face recognizer used in Pi deployment.

1.2 Dataset Structure

Create the following folder structure in your project:

```
face_voice_project/
  dataset/
    faces/
      Bakiya/     (10 photos each)
      Ranjani/
      srikavi/
      Swetha/
    voices/
      Bakiya/     (10 .wav clips each)
      Ranjani/
      srikavi/
      Swetha/
```

1.3 Train Voice Model (train_voice.py)

Features extracted per voice clip:

- MFCC - 40 coefficients (mean + std)
- Pitch features (mean, std)
- Energy/RMS (mean, std)
- Spectral centroid and rolloff
- Total: 86 features per clip

Model pipeline:

```
StandardScaler -> SVM(kernel='rbf', C=10, gamma='scale')
Output: voice_model.pkl
Accuracy: 100% on training set
```

1.4 Export Face Model for Pi (export_for_pi_v2.py)

Creates a lightweight LBPH face model that works without TensorFlow on Pi:

```
python export_for_pi_v2.py

Output files:
  lbph_face_model.yml     <- LBPH trained model
  lbph_label_names.pkl    <- Person name labels
```

NOTE: LBPH (Local Binary Pattern Histogram) is OpenCV's built-in face recognizer. Much faster than DeepFace on Pi.

SECTION 2: Raspberry Pi 4B Setup

2.1 Install System Packages

```
sudo apt update && sudo apt upgrade -y

sudo apt install -y python3-pip python3-venv \
  portaudio19-dev libgl1 libsm6 libxext6 \
  libjpeg-dev libopenblas-dev liblapack-dev \
  python3-opencv libgtk-3-dev libgtk2.0-dev \
```

```
  pkg-config i2c-tools python3-tk
```

2.2 Create Virtual Environment

```
mkdir ~/face_voice_project
cd ~/face_voice_project
python3 -m venv env
source env/bin/activate
```

2.3 Install Python Packages on Pi

```
pip install --upgrade pip
pip install numpy scikit-learn sounddevice librosa pillow
pip install opencv-python
pip install opencv-contrib-python
pip install smbus2 luma.oled RPi.GPIO
pip install pyserial pynmea2 scipy
```

2.4 Transfer Files from PC to Pi

Run these commands in PyCharm terminal on Windows PC:

```
# Transfer model files
scp lbph_face_model.yml om@10.77.154.58:/home/om/face_voice_project/
scp lbph_label_names.pkl om@10.77.154.58:/home/om/face_voice_project/
scp voice_model.pkl om@10.77.154.58:/home/om/face_voice_project/

# Transfer scripts
scp recognize_pi.py om@10.77.154.58:/home/om/face_voice_project/
scp health_monitor.py om@10.77.154.58:/home/om/face_voice_project/
scp dashboard.py om@10.77.154.58:/home/om/face_voice_project/

# Transfer dataset (entire folder)
scp -r dataset om@10.77.154.58:/home/om/face_voice_project/
```

SECTION 3: Hardware Wiring

3.1 MAX30102 Heart Rate Sensor (I2C)

MAX30102 Pin	Pi Pin Number	Pi GPIO Name	Notes
VCC	Pin 1	3.3V	Use 3.3V only
GND	Pin 6	GND	
SDA	Pin 3	GPIO2 (SDA)	I2C Data
SCL	Pin 5	GPIO3 (SCL)	I2C Clock

I2C Address: 0x57

3.2 OLED Display 1.3 inch (I2C)

OLED Pin	Pi Pin Number	Pi GPIO Name	Notes
VCC	Pin 1	3.3V	Use 3.3V only
GND	Pin 6	GND	
SDA	Pin 3	GPIO2 (SDA)	Same as MAX30102
SCL	Pin 5	GPIO3 (SCL)	Same as MAX30102

I2C Address: 0x3C

NOTE: Both MAX30102 and OLED share the same I2C pins. Use a small breadboard to connect both sensors to the same pins.

3.3 NEO-M8N GPS Module (UART)

GPS Pin	Pi Pin Number	Pi GPIO Name	Notes
VCC	Pin 2	5V	Must use 5V
GND	Pin 14	GND	
TX	Pin 10	GPIO15 (RXD)	GPS TX goes to Pi RX
RX	Pin 8	GPIO14 (TXD)	GPS RX goes to Pi TX

NOTE: TX and RX always cross over - GPS TX connects to Pi RX and vice versa.

3.4 Enable I2C and UART

Enable I2C:

```
sudo raspi-config
-> Interface Options -> I2C -> Yes
```

Enable UART:

```
sudo raspi-config
-> Interface Options -> Serial Port
-> Login shell over serial: No
-> Serial hardware enabled: Yes
```

Edit /boot/firmware/config.txt and add at bottom under [all]:

```
enable_uart=1
dtoverlay=miniuart-bt
```

Disable Bluetooth service:

```
sudo systemctl disable bluetooth
sudo reboot
```

3.5 Verify Sensors are Detected

```
# Check I2C devices
i2cdetect -y 1

# Expected output:
# 0x3C  <- OLED display
# 0x57  <- MAX30102 sensor

# Check GPS serial port
ls -la /dev/serial*
# Expected: serial0 -> ttyAMA0

# Test GPS data
sudo cat /dev/serial0
```

SECTION 4: Face & Voice Recognition

4.1 How It Works

The recognition system uses a 5-state machine:

State	Description
IDLE	Press SPACE to start
FACE_SCAN	10 second countdown - look at camera
FACE_RESULT	Shows face match result - press SPACE to continue
VOICE_SCAN	5 second recording - speak clearly
RESULT	Shows match/no match - auto resets after 8 seconds

4.2 Face Recognition Method

- Uses LBPH (Local Binary Pattern Histogram) - OpenCV built-in
- No TensorFlow needed on Pi - very fast
- Training done on PC, model transferred to Pi
- Confidence threshold: < 100 = recognized
- Lower confidence score = better match

4.3 Voice Recognition Method

- Records 5 seconds of audio at 16000 Hz
- Extracts 86 features per recording
- Uses SVM classifier with rbf kernel
- Microphone: USB Camera built-in mic (device 1, stereo)
- Stereo audio converted to mono before processing

4.4 Run Face + Voice Recognition

```
cd ~/face_voice_project
source env/bin/activate
python3 recognize_pi.py
```

SECTION 5: Health Monitor (MAX30102 + OLED)

5.1 What It Measures

Measurement	Method	Normal Range
Heart Rate	FFT frequency analysis on IR signal	60-100 BPM
SpO2	Ratio of Red/IR light absorption	95-100%

5.2 OLED Display Layout

```
+------------------+
| HEALTH MONITOR   |
|------------------|
| Heart Rate:      |
| 75 BPM           |
|------------------|
| SpO2:    98 %    |
+------------------+
```

5.3 Tips for Accurate Readings

1. Place finger gently on sensor - do not press hard
2. Keep finger completely still during measurement
3. Cover the sensor completely with fingertip
4. Wait 10 seconds after placing finger
5. Keep still until reading appears on OLED

5.4 Run Health Monitor

```
cd ~/face_voice_project
source env/bin/activate
python3 health_monitor.py
```

SECTION 6: GPS Tracking (NEO-M8N)

6.1 GPS Features

- Tracks location continuously
- Sets home location on first GPS fix
- Calculates distance from home using Haversine formula
- Shows alert on screen when person moves beyond 10 meters
- Displays latitude, longitude and satellite count

6.2 GPS Fix Tips

- Place GPS antenna near a window
- Antenna ceramic side must face upward toward sky
- First fix takes 1-2 minutes
- Subsequent fixes are much faster

6.3 Serial Port Configuration

```
# GPS uses /dev/serial0 which maps to /dev/ttyAMA0
# Baud rate: 9600

# Test GPS data:
sudo cat /dev/serial0

# Should see NMEA sentences like:
# $GNGGA,123456,1234.5678,N,08012.3456,E,...
```

SECTION 7: Complete Dashboard (dashboard.py)

7.1 Dashboard Layout

```
+-----------------------------------------------------+
| WARNING: Person moved beyond 10m! (red banner)      |
+-----------------+----------+----------------------+
|                 | Heart    | SpO2                 |
| Live Camera     | Rate     |                      |
| Feed            | 75 BPM   | 98 %                 |
|                 +----------+----------------------+
|                 | GPS      | Face + Voice         |
|                 | Location | Recognition Buttons  |
+-----------------+----------+----------------------+
| Status Bar - current system status                  |
+-----------------------------------------------------+
```

7.2 Dashboard Features

Feature	Description
Live Camera	Shows webcam feed with face detection box
Heart Rate Card	Shows BPM from MAX30102 sensor
SpO2 Card	Shows blood oxygen percentage
GPS Card	Shows coordinates, distance from home, satellites
Recognition Card	Face Scan and Voice Scan buttons
Alert Banner	Red banner appears when person moves beyond 10m
Set Home Button	Updates home location to current GPS position

7.3 Run Dashboard

```
cd ~/face_voice_project
source env/bin/activate
python3 dashboard.py
```

SECTION 8: Troubleshooting

8.1 Common Issues and Fixes

Problem	Cause	Fix
Camera black screen	Wrong device or needs warmup	Use cv2.CAP_V4L2, read 10 frames before using
Face shows Unknown	Histogram method too simple	Use LBPH model from export_for_pi_v2.py
Voice mic error	Wrong channels (1 vs 2)	Set channels=2 then convert to mono
GPS no data	BT using UART	Add dtoverlay=miniuart-bt to config.txt
WiFi stopped	disable-bt breaks WiFi	Replace with dtoverlay=miniuart-bt
OLED blank	Wrong display driver	Use sh1106 not ssd1306 for 1.3 inch
I2C not found	I2C not enabled	Run raspi-config, enable I2C
cv2.face error	Need contrib package	pip install opencv-contrib-python

8.2 SD Card Editing on Windows

If Pi cannot connect to WiFi and you need to edit config.txt:

6. Remove SD card from Pi
7. Insert into Windows PC
8. Download DiskGenius (free) from diskgenius.com
9. Open DiskGenius, find bootfs partition
10. Right click config.txt, select Copy To Desktop
11. Edit in Notepad, make changes
12. Back in DiskGenius, right click empty space
13. Click Copy Files To Current Partition, select Overwrite
14. Click Save All in DiskGenius
15. Safely eject SD card and put back in Pi

8.3 Useful Test Commands

```
# Test all Python imports
python3 -c "import cv2, numpy, sklearn, librosa, sounddevice; print('All OK')"

# List cameras
v4l2-ctl --list-devices

# List audio devices
python3 -c "import sounddevice; print(sounddevice.query_devices())"

# Check microphone
arecord -D hw:3,0 --dump-hw-params -d 1 -f S16_LE -r 16000 -c 2 test.wav

# Check I2C sensors
i2cdetect -y 1

# Check GPS port
ls -la /dev/serial*
sudo cat /dev/serial0
```

SECTION 9: Project File Reference

9.1 Windows PC Files

File	Purpose
train_voice.py	Train voice recognition SVM model
export_for_pi_v2.py	Create LBPH face model for Pi
recognize.py	Full recognition script for Windows
voice_model.pkl	Trained voice SVM model
lbph_face_model.yml	Trained LBPH face model

File	Purpose
lbph_label_names.pkl	Person name labels

9.2 Raspberry Pi Files

File	Purpose
recognize_pi.py	Face + Voice recognition (no GUI)
health_monitor.py	MAX30102 + OLED health display
dashboard.py	Full GUI dashboard - all features
test_oled.py	Test OLED display
test_sensor.py	Test MAX30102 sensor
test_gps.py	Test GPS module
lbph_face_model.yml	Transferred from PC
lbph_label_names.pkl	Transferred from PC
voice_model.pkl	Transferred from PC
dataset/	Face photos and voice clips

9.3 Running Both Scripts Together

Open two terminals on Pi and run one script in each:

```
# Terminal 1 - Face and Voice Recognition
cd ~/face_voice_project && source env/bin/activate
python3 recognize_pi.py

# Terminal 2 - Health Monitor (OLED)
cd ~/face_voice_project && source env/bin/activate
python3 health_monitor.py

# OR run everything in one dashboard:
python3 dashboard.py
```

End of Guide - Project Complete